

Various software are available for using the FM receiver in Windows OS, but these do not provide options for research work as available in GNU Radio. Windows-based GNU Radio is

available, but it requires manual driver installation for hardware.

The live disk image based on Ubuntu with GNU Radio Companion can be downloaded from <https://wiki.gnuradio.org/index.php/installingGR>

QR code for the website is given in Fig. 2. Once the file is downloaded, you can easily create a DVD with GNU Radio Boot Support using any DVD burning software. During testing, Etcher software was used for burning the live disk image bootable DVD.

Working on GNU Radio

Working on GNU Radio

Boot using CD/DVD drive or USB drive as per your bootable media. The process will take around five minutes to complete initial setup. Now, use GNU Radio for signal processing applications. GNU Radio icon will appear on the desktop, as shown in Fig. 3.

Test RTL-SDR support in GNU Radio live disk environment. Check proper working of RTL-SDR dongle with the system. For that, add RTL-SDR block from the library of tool box in GNU Radio, and a waterfall block or scope block (steps for designing are given in the next section). Then, provide centre frequency in RTL-SDR block. Insert the dongle in your computer and run the flowgraph (or program) in GNU Radio. Test flow chart is shown in Fig. 4.

The antenna is a standard antenna provided with RTL-SDR dongle with a magnetic base, as shown in Fig. 5. It provides acceptable FM reception in frequency range of 88MHz to 108MHz. Place the antenna at a proper place. If you place it close to the chargers, cellphones or electric sockets, some disturbances may be detected while testing the receiver.

Steps for designing FM receiver in GNU Radio

Generate, execute and kill are three important tools in GNU Radio software. The program flowgraph consists of various blocks.

RTL-SDR block as signal source. Add RTL-SDR source

TABLE I
RTL-SDR BLOCK
PARAMETER VALUES

Parameter	Value
Sample rate (sps)	2 mega samples per second
Ch0: Frequency (Hz)	98.3MHz (can be controlled by slider for tuning)
Ch0: Frequency correction (ppm)	0
Ch0: DC offset mode	Off
Ch0: IQ balance mode	Off
Ch0: Gain mode	Manual
Ch0: RF gain (db)	10
Ch0: IF gain (db)	20
Ch0: BB gain (db)	20

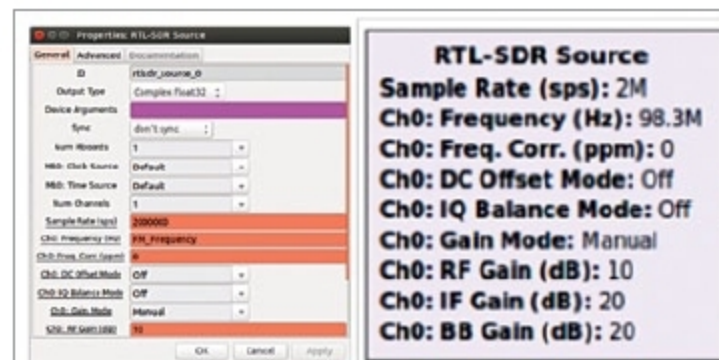


Fig. 6: RTL-SDR block settings window

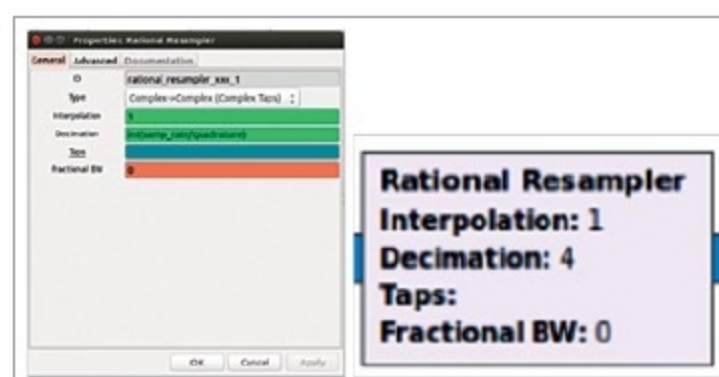


Fig. 7: Rational resampler block settings with 1 interpolation

TABLE II
RATIONAL RESAMPLER
VALUES

Parameter	Value
Interpolation	1
Decimation	4
Taps	-
Fractional bandwidth	0

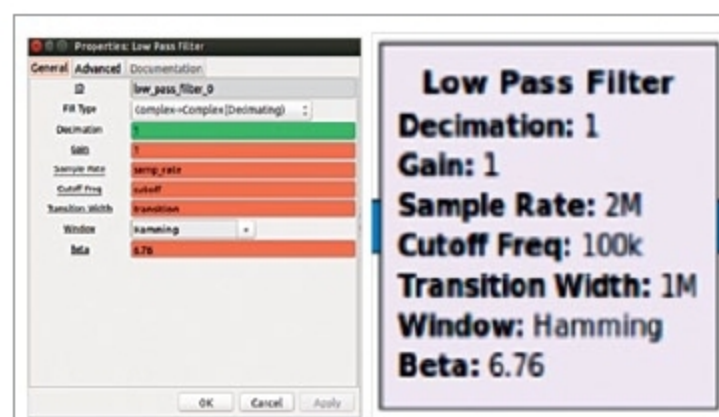


Fig. 8: Low-pass filter block settings

TABLE III
LOW-PASS FILTER VALUES

Parameter	Value
Decimation	1
Gain	1
Sample rate	2Msps
Cut-off frequency (Hz)	100k
Transition width	1MHz
Window	Hamming
Beta	6.76

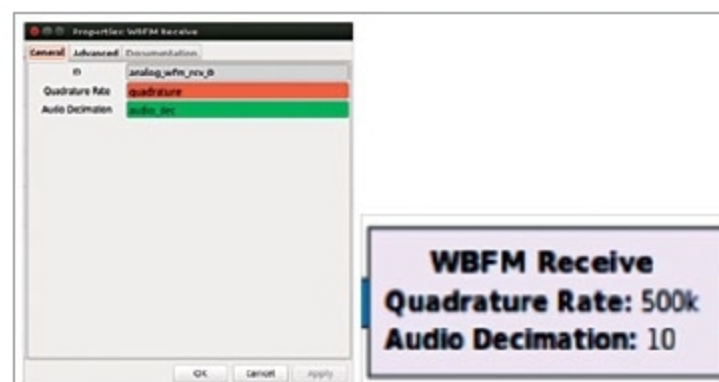


Fig. 9: WBFM block

TABLE IV
WBFM BLOCK PARAMETERS

Parameter	Value
Quadrature rate	500k
Audio decimation	10

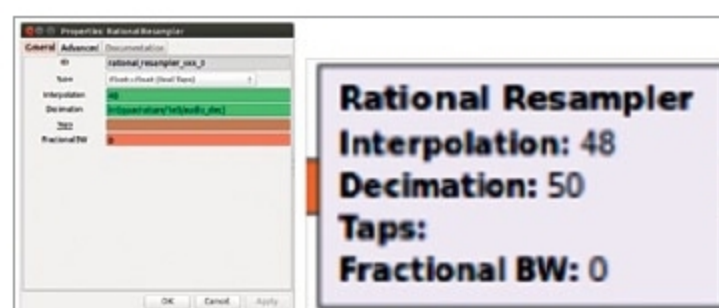


Fig. 10: Rational resampler block with 48 interpolation